Better Software for Reproducible Science

Presented by

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Members of

Consortium for the Advancement of Scientific Software
https://cass.community

With prior support from

David E. Bernholdt, Anshu Dubey

The International Conference for High-Performance Computing, Networking, Storage, and Analysis (SC25) conference

Slides, late-breaking updates, etc. at:
https://bssw-tutorial.github.io/
and click the link for today's tutorial

License, Citation and Acknowledgements

License and Citation

- **The requested citation the overall tutorial is:** David E. Bernholdt and Anshu Dubey, Better Software for Reproducible Science tutorial, in The International Conference for High-Performance Computing, Networking, Storage, and Analysis (SC25), St. Louis, Missouri, 2025. DOI: 10.6084/m9.figshare.30394186.
- Individual modules may be cited as *Speaker, Module Title*, in *Tutorial Title*, …

Acknowledgements

- This work was supported by the U.S. Department of Energy, Office of Science, Office of Advanced Scientific Computing Research, Next-Generation Scientific Software Technologies (NGSST) and Scientific Discovery through Advanced Computing (SciDAC) programs.
- This work was supported by the U.S. Department of Energy Office of Science, Office of Advanced Scientific Computing Research (ASCR), and by the Exascale Computing Project (17-SC-20-SC), a collaborative effort of the U.S. Department of Energy Office of Science and the National Nuclear Security Administration.
- This work was performed in part at the Argonne National Laboratory, which is managed by UChicago Argonne, LLC for the U.S. Department of Energy under Contract No. DE-AC02-06CH11357.
- This work was performed in part at the Lawrence Livermore National Laboratory, which is managed by Lawrence Livermore National Security, LLC for the U.S. Department of Energy under Contract No. DE-AC52-07NA27344.
- This work was performed in part at the Los Alamos National Laboratory, which is managed by Triad National Security, LLC for the U.S. Department of Energy under Contract No.89233218CNA000001
- This work was performed in part at the Oak Ridge National Laboratory, which is managed by UT-Battelle, LLC for the U.S. Department of Energy under Contract No. DE-AC05-00OR22725.
- This work was performed in part at Sandia National Laboratories. Sandia National Laboratories is a multi-mission laboratory managed and operated by National Technology and Engineering Solutions of Sandia, LLC., a wholly owned subsidiary of Honeywell International, Inc., for the U.S. Department of Energy's National Nuclear Security Administration under contract DE-NA0003525.

About Us

- David E. Bernholdt, ORNL
- Anshu Dubey, ANL

David
he/him

Anshu
she/her

- Participants in the Consortium for the Advancement of Scientific Software (CASS) and DOE/ASCR software stewardship projects: https://cass.community/
  - Members of the PESO and RAPIDS projects
- Past involvement in
  - Exascale Computing Project
  - IDEAS Productivity family of projects: https://ideas-productivity.org/
  - And more!

Building an Online Community

https://bssw.io

- **New community-based resource for scientific software improvement**
- A central hub for sharing information on practices, techniques, experiences, and tools to improve developer productivity and software sustainability for computational science & engineering (CSE)

Goals

- Raise awareness of the importance of **good software practices** to scientific productivity and to the quality and reliability of computationally-based scientific results
- Raise awareness of the **increasing challenges** facing CSE software developers as high-end computing heads to extreme scales
- Help CSE researchers **increase effectiveness** as well as leverage and impact
- **Facilitate CSE collaboration via software** in order to advance scientific discoveries

Site users can…

- **Find information** on scientific software topics
- **Contribute new resources** based on your experiences
- Create content tailored to the unique needs and perspectives of a focused scientific domain

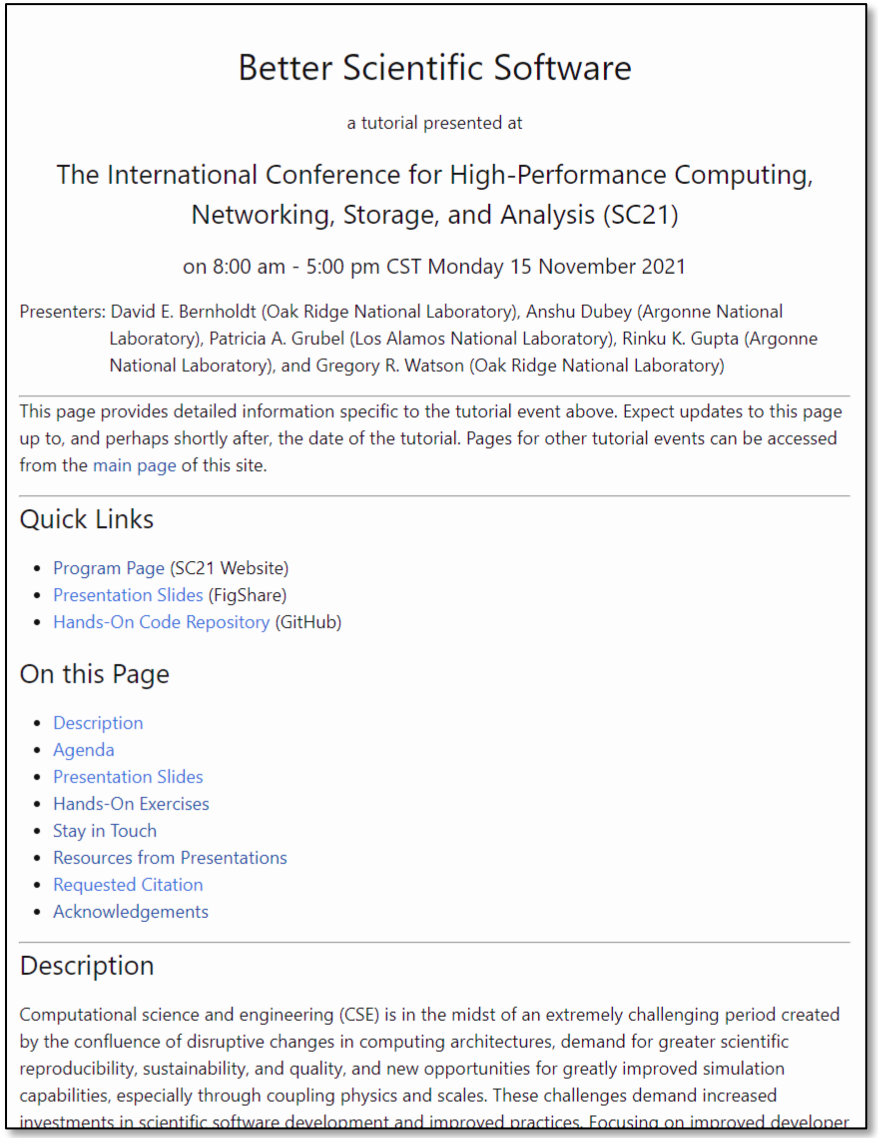
Follow BSSw

- BSSw Digest: https://bssw.io/pages/receive-our-email-digest
  - Updates on BSSw content
    - New blog posts, events, and resources
    - BSSw Fellowship
  - Typically 1-2 messages per month
  - Also: RSS feed: https://bssw.io/items.rss

better scientific software

BSSw Tutorial Web Site

- https://bssw-tutorial.github.io/ is the one URL you need to find all the resources for this tutorial
- Each tutorial event has its own page
- Each tutorial page is considered archival
  - All of the materials used in that tutorial (or links to them)
  - Materials may be updated if we find mistakes



Better Scientific Software

a tutorial presented at

The International Conference for High-Performance Computing, Networking, Storage, and Analysis (SC21)

on 8:00 am - 5:00 pm CST Monday 15 November 2021

Presenters: David E. Bernholdt (Oak Ridge National Laboratory), Anshu Dubey (Argonne National Laboratory), Patricia A. Grubel (Los Alamos National Laboratory), Rinku K. Gupta (Argonne National Laboratory), and Gregory R. Watson (Oak Ridge National Laboratory)

This page provides detailed information specific to the tutorial event above. Expect updates to this page up to, and perhaps shortly after, the date of the tutorial. Pages for other tutorial events can be accessed from the main page of this site.

Quick Links

- Program Page (SC21 Website)
- Presentation Slides (FigShare)
- Hands-On Code Repository (GitHub)

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- Acknowledgements

Description

Computational science and engineering (CSE) is in the midst of an extremely challenging period created by the confluence of disruptive changes in computing architectures, demand for greater scientific reproducibility, sustainability, and quality, and new opportunities for greatly improved simulation capabilities, especially through coupling physics and scales. These challenges demand increased

Explaining Slide 2

- Slide 2 in all of our presentations contains the license, citation, and acknowledgements for the tutorial
- (Software) best practice to make your license and preferred citation(s) easily findable
- Sponsor acknowledgements rarely hurt!

License, Citation and Acknowledgements

License and Citation

- **The requested citation the overall tutorial is:** Anshu Dubey, David E. Bernholdt, Better Scientific Software tutorial, in ISC High Performance, Hamburg, Germany and online, 2024

Acknowledgements

- Material included in these presentation is derived from work supported by the U.S. Department of Energy Office of Science, Office of Advanced Scientific Computing Research (ASCR), and by the Exascale Computing Project (17-SC-20-SC), a collaborative effort of the U.S. Department of Energy Office of Science and the National Nuclear Security Administration.
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- This work was performed in part at the Oak Ridge National Laboratory, which is managed by UT-Battelle, LLC for the U.S. Department of Energy under Contract No. DE-AC05-00OR22725.

We Want to Interact with You!

- We find these tutorials most interesting and informative (for everyone) if you ask questions and share experiences!
  – We learn too!
- **In the room**: please raise your hand at any time to ask a question, but wait for the mic so that everyone in the room and online will be able to hear you
- **Online**: please use the Digital Experience Q&A tool to ask your question. We'll respond in the Q&A too or verbally as opportunities permit. Please remember that the online stream is delayed by about 15 seconds, so there will be some latency in our responses
- After the tutorial email us at bssw-tutorial@lists.mcs.anl.gov
  – With questions or feedback
  – The list moderator will allow your messages to be posted
- Refer to bssw-tutorial.github.io page for all tutorial materials

You may also be interested in these other software-related events at SC25: https://bssw.io/events/sc25-software-related-events
(link is also on tutorial web page)

Hands-On Activities

- **For detailed information, see the tutorial web page: https://bssw-tutorial.github.io/2025-11-16-sc/#hands-on-activities**
- The “Responsible Software Development with LLMs” module includes a hands-on component
  - You will be using an LLM to generate code and tests
- Participation in the hands-on is encouraged but not required
  - The instructor’s prompts and generated code will be made available to participants after the exercise
- **To participate in the hands-on, you will need access to a web-based LLM chat tool**
  - The instructor will be using ChatGPT, but any similar tool will work
  - This is the “**web interface**” path in the instructions on the website

Agenda

The agenda is also available on the tutorial web page. Visit https://bssw-tutorial.github.io and click on the link for today's tutorial

Time (CST)	Title	Presenter
8:30 AM	Introduction	David E. Bernholdt (ORNL)
8:45 AM	Motivation and Overview of Best Practices in HPC Software Development	David E. Bernholdt (ORNL)
9:15 AM	Improving Reproducibility Through Better Software Practices	David E. Bernholdt (ORNL)
10:00 AM	*Morning break*	
10:30 AM	Responsible Software Development with LLMs	Anshu Dubey (ANL)
12:00 PM	*Adjourn*	